

ROSHAN PRABHU

AI / ML ENGINEER | GENERATIVE AI | FULL-STACK & AUTOMATION

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PROFESSIONAL SUMMARY

AI/ML and software developer with hands-on experience building AI-powered applications, LLM workflows, document intelligence solutions, voice agents, and full-stack business systems. Skilled in Python, JavaScript, Node.js, React, machine learning, prompt engineering, API integration, and cloud deployment. Experienced in turning business requirements into usable software, debugging production issues, and integrating AI with validation, explainability, and human-review controls.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, C, Kotlin, SQL

AI / Machine Learning: Generative AI, LLM Integration, Prompt Engineering, Computer Vision / Vision-Language Models, OCR, Model Training & Deployment, scikit-learn, Predictive Automation

Frameworks & Libraries: LangChain, LangGraph, Node.js, Express.js, React, Vite, FastAPI, Streamlit, Pandas, NumPy

Databases & Data: MongoDB, MongoDB Atlas, PostgreSQL, Supabase, Pinecone, SQL, Data Preprocessing

Cloud & Developer Tools: AWS EC2, AWS S3, Railway, Netlify, Docker, Git, GitHub, PM2, Nginx

APIs & Integrations: OpenAI API, Gemini API, Meta Marketing API, WhatsApp Business API, MSG91, Cloudinary, Razorpay, Telephony APIs

Analytics & Productivity: Matplotlib, Chart.js, Excel, REST APIs, Dashboard Development

PROFESSIONAL EXPERIENCE

Freelance & Internship Projects — AI & Web Development | AI-Powered Application Developer

2023 – Present

- Designed and developed AI-powered web applications, automation tools, dashboards, and backend services using Python and JavaScript.
- Built an AI Insurance Advisor chatbot that provides personalized, profile-based recommendations supported by rule-based reasoning for more explainable outputs.
- Developed secure chat interfaces and dashboards, integrated third-party APIs, and supported cloud deployment and Git-based version control.
- Contributed to end-to-end software delivery, including requirements translation, implementation, debugging, integration, and deployment support.

Aixplora Technologies Pvt. Ltd. | AI Engineer Intern

July 2025 – November 2025

- Developed an AI-driven financial planning system for personalized term-insurance recommendations.
- Designed profile-based analysis and decision logic to support recommendation relevance and consistency.
- Contributed to explainable AI decision-making for financial-advisory use cases.

SELECTED AI / ML & SOFTWARE PROJECTS

Saanvi — Real-Time Multilingual AI Voice Agent

Technologies: Gemini Live, GPT-4.1-nano, Deepgram, ElevenLabs, telephony APIs, Node.js

- Built a real-time phone-agent pipeline combining speech-to-text, LLM response generation, and text-to-speech for lead qualification and callback workflows.
- Reduced measured LLM first-token latency from approximately 1.2 seconds to 0.4 seconds through model selection and generation-parameter tuning; project notes report sub-second conversational turn latency.
- Debugged WebSocket setup sequencing, audio framing, and sample-rate conversion issues in telephony audio streaming.

AI Marksheet & Document Scanner

Technologies: Kotlin, CameraX, Node.js, Gemini, OpenAI, Azure Document Intelligence

- Developed an Android and backend application to extract marks and structured data from photographed handwritten documents, with export workflows for Excel, CSV, PDF, and Word.
- Designed template-aware, two-pass extraction and arithmetic validation to flag inconsistent subtotals and totals for human review.
- Added backend-proxied model requests, user usage limits, credit billing, push notifications, and administrative access controls.

AI Insurance Advisor Chatbot

Technologies: Python, LangChain, Pinecone, Hugging Face embeddings, LangGraph, rule-based reasoning

- Built a conversational insurance-advisory application that uses user-profile information and insurance knowledge to generate personalized product recommendations.
- Combined retrieval-based knowledge access with deterministic rules to improve consistency and make recommendation logic more explainable.

Factory Management & Production Tracking System

Technologies: React, Vite, Node.js, Express, MongoDB, AWS EC2, S3

- Contributed deployment and debugging for a live production-management PWA with purchase-order and photo-upload workflows and administrative/merchandise views.
- Resolved a frontend build-output and Nginx serving-path mismatch and established a build-and-copy deployment sequence on AWS.

CRM Lead Nurturing & WhatsApp Automation

Technologies: Node.js, MongoDB, MSG91, WhatsApp Business API

- Implemented five lead-nurture tracks using scheduled jobs and lead-status event triggers, with messaging restricted to configured business hours.
- Implemented message-template handling and personalization and troubleshoot WhatsApp template-format and approval issues.

Automated Advertising Performance Reporting

Technologies: Meta Marketing API, Supabase/PostgreSQL, Node.js, React, GPT-4o-mini

- Built a reporting workflow that collects advertising metrics, stores campaign data, generates AI-assisted insights, and delivers daily and weekly PDF reports and alerts.
- Fixed stale-spend field mapping and weekly aggregation issues and validated automated report delivery.

Server-Rendered Marketing Website & Performance Optimization

Technologies: React 19, Vite SSR, Three.js, GSAP, Lenis, Tailwind CSS

- Built and optimized a server-rendered marketing website with 3D scenes and scroll-driven animation.
- Removed a large wildcard icon import, resolved animation pin conflicts, and addressed mobile scroll-lock and route-rendering issues.

Hospital Revenue-Share Settlement Ledger

Technologies: JavaScript, Netlify Functions, MongoDB Atlas

- Converted spreadsheet-based doctor revenue-share, TDS, and hospital settlement rules into a web ledger backed by serverless functions and a database.

SDN Network Traffic Classification

Technologies: Python, scikit-learn, Mininet, Ryu

- Developed a Software-Defined Networking traffic-classification project using a Decision Tree model in a Mininet/Ryu test environment.

AI-Powered Smart Sewage Treatment System (In Development)

Technologies: Python, Streamlit, Machine Learning, ESP32 / IoT integration

- Developing a water-quality monitoring and prediction dashboard with sensor integration, BOD/COD estimation, alerts, and treatment-control logic for a 100 KLD STP context.

Togvel — Group Travel Platform

Technologies: Web application development, workflow design, analytics

- Developed a group-travel platform to simplify travel planning, including secure workflows, administrative controls, and analytics.

Courier Management System

Technologies: Web application development, parcel tracking

- Created a courier operations system for parcel tracking and management workflows.

EDUCATION

Impact College of Engineering and Applied Science

Bachelor of Engineering (B.E.), Artificial Intelligence & Machine Learning | Dec 2021 – Jul 2025 | CGPA: 7.43 / 10

- Awarded Best Outgoing Student for academic excellence, leadership, and overall contribution.
- Volunteered at the college annual technical fest, coordinating event logistics and registrations.

ADDITIONAL INFORMATION

GitHub: github.com/29roshan-prog